PROJECT CANARY · TEAM STUDY GUIDE

Capstone Defense Cheat Sheet

What Canary is, how each component works, what the results mean, and which claims the evidence supports

Core story: Identify buildings that need attention, project Day 35 weight and harvest recovery, explain the warning signs, and recommend the next management check.

1. The 20-second answer

Project Canary is an early-warning and decision-support system for a broiler farm. It combines daily farm records into one view per building, assigns an explainable operational risk rating, forecasts Day 35 average weight and final harvest recovery, and recommends what management should inspect next.

The two farm goals are:

- **1,800 g average weight on Day 35**
- **95% harvest recovery**, defined for this capstone as ending or surviving birds divided by beginning population

Days 1–14 are the early-warning window. Monitoring continues through the active cycle.

2. Business question

Which buildings are at risk, what Day 35 weight and harvest-recovery outcomes are currently expected, why are they at risk, and what should management check next?

3. The three components

Component	What it answers	Output
Rules-based risk scoring	Which building needs attention and why?	Low, Medium, High, or Critical; 0–12 score; dimension evidence
Predictive models	What outcome is likely from what is known today?	Projected Day 35 average weight and projected final recovery, with ranges
Recommendation playbook	What should management inspect next?	Pattern-based action, urgency, checklist, and escalation trigger

Important: the risk score and predictions are separate. A forecast does not add points to the risk score.

4. Data preparation

One reliable building-day table

- Source rows: **1,785**
- Unique building-day rows: **1,666**
- Repeated building-days consolidated: **119**
- Production-value conflicts in repeated rows: **0**
- Recorded weight-measurement days: **210**

The canonical key is:

Harvest cycle + building + production day

Zone A and Zone B

In 2026-2 and 2026-3, some building-days have one environmental row for Zone A and another for Zone B. Canary does **not** treat these as two independent flock observations.

For each affected building-day:

- average temperature/humidity = unweighted average of the two zone averages
- minimum = minimum across zones
- maximum = maximum across zones
- zone spread = absolute difference between zone averages
- production fields such as population, mortality, feed, and weight are retained once

Why unweighted? Zone-level flock counts or floor-area weights are unavailable. This is transparent and provisional.

Corrected weight records

The updated workbook now contains the corrected checkpoint weights:

- 2026-1: Days 7, 14, 21, 28, and 35 for all six buildings
- 2026-2: Days 7, 14, 21, 28, and 35 for all six buildings
- 2026-3: Days 7, 14, 21, 28, and 35 for Tags 1–3

The latest cycle, 2026-3, is excluded from model training so current/future information cannot leak into validation.

5. Revised target-weight curve

Doc Raymond's approved checkpoints are:

Day	Target
7	170 g
14	380 g
21	800 g
28	1,200 g
35	1,800 g

Canary creates two daily curves:

- **Linear:** spreads each weekly gain evenly across seven days.
- **Smoothed, used by Canary:** preserves the previous farm curve's proportional within-week growth shape, then rescales the segment to hit the revised weekly endpoints exactly.

The Day 0 anchor remains a **40 g working assumption from the former curve**; it was not part of Doc Raymond's revised checkpoint list. Targets stay at 1,800 g after Day 35 for milestone comparisons.

The smoothed curve is used for:

- the rules-based weight-gap calculation
- current-weight-to-age-target progress in the Day 35 model
- dashboard target charts

It is **not** the model target. The model target is the actual recorded Day 35 weight.

6. Component 1 — Risk scoring

Four dimensions

Each dimension scores 0–3 points.

Dimension	Question	Evidence
Weight gap	How far below the smoothed target was the latest measured weight at that measurement age?	Actual weight versus target for the weighing day
Survival path	Is the surviving share below the expected path to 95% by Day 35?	Current survival versus age-adjusted expected survival
Mortality trend	Is recent mortality worsening versus the recent baseline?	Last 3 days versus prior 7-day baseline
Peer comparison	Is this building worse than age-matched buildings in the same cycle?	Weight, survival, and mortality differences versus peers

Weight-gap thresholds

Gap below age target	Score
Under 5%	0
5% to under 15%	1
15% to under 30%	2
30% or more	3

Survival and mortality thresholds are age-aware. Peer thresholds require at least three comparable buildings within two production days.

Total score to label

Total	Label
0–1	Low
2–3	Medium
4–5	High
6–12	Critical

Why environment is not a fifth score

Temperature, humidity, feed, and water are potential **causes or investigation signals**, while weight, survival, and mortality are observed performance outcomes. Adding every environmental signal to the 0–12 score would risk double-counting and make the score harder to defend.

Canary therefore uses a two-level design:

1. The four-dimension score prioritizes buildings.
2. Operational alerts investigate likely causes using temperature, humidity, feed, mortality, and data freshness.

Water and THI are deferred until reliable building-day data and farm-approved thresholds exist.

Honest limitation

Risk thresholds are provisional until Doc Raymond signs them off. The score represents operational concern, not a statistical probability of missing a goal.

7. Component 2A — Day 35 weight model

Business question and target

Question: Given the checkpoint weights known today, what average building weight should we expect on Day 35?

Y: observed building average bodyweight on production Day 35.

Training unit

- **31 independent building-cycle Day 35 outcomes**
- **6 historical cycles:** 2025-2 through 2026-2
- **124 as-of checkpoint rows:** 31 outcomes × Days 7, 14, 21, and 28

The 124 rows are not 124 independent final outcomes. They are four decision snapshots for each of 31 outcomes.

Leakage-safe inputs

At each checkpoint, only weights already known are included:

- Day 7 row: Day 7 weight only

- Day 14 row: Days 7 and 14
- Day 21 row: Days 7, 14, and 21
- Day 28 row: Days 7, 14, 21, and 28

Additional inputs:

- current measurement day
- current measured weight
- current weight divided by the smoothed target for that age
- recent average daily gain when a previous measurement exists
- cumulative gain from Day 7
- indicators for whether growth-history inputs are available

Models compared

Candidate	Held-out MAE	Result
Historical Day 35 mean	about 210 g	Baseline
Target-curve ratio	about 295 g	Not selected
Recent linear ADG	about 432 g	Not selected
Historical remaining gain	about 178 g	Strong transparent benchmark
Ridge regression	**about 172 g**	**Selected**
Random Forest	about 176 g	Not selected
Gradient boosting	about 203 g	Not selected

Validation and selection

Canary uses leave-one-cycle-out cross-validation. Every row from a held-out cycle stays out of training.

Primary selection metric: **cycle-balanced MAE in kilograms**, reported to the business in grams.

Rule: choose the simplest candidate within 5% of the best cycle-balanced MAE. Historical remaining gain was more than 5% worse than Ridge, so Ridge became the champion.

Selected-model results

- Overall MAE: **about 172 g**
- RMSE: **about 232 g**
- Within 200 g: **about 65%**
- Correct side of the revised 1.8 kg target: **about 86%**
- Day 14 MAE: **about 167 g**
- Day 14 correct target side: **about 87%**

There are only **5 historical 1.8 kg hits** and 26 misses. The model catches misses well but recognizes only about 20% of the small hit group. Do not oversell target-hit classification.

Feature importance

Ridge importance is based on absolute standardized coefficients. The leading fitted inputs are target progress, Day 14 weight, recent gain, Day 21 weight, and cumulative gain.

These are model associations—not proof that changing one input causes the final result. Correlated growth variables can share importance or have counterintuitive signs.

8. Historical average remaining gain — full example

This method is still important because it is the strongest simple benchmark.

How it is computed

For every eligible historical building-cycle at the same checkpoint:

Remaining gain = actual Day 35 weight − checkpoint weight

At Day 14, Canary averages this across the 31 historical building outcomes:

Average Day 14-to-35 remaining gain = 1.254981 kg ≈ 1,255 g

End-to-end illustrative forecast

Suppose a current building weighs **380 g on Day 14**.

1. Current measured weight = 380 g.
2. Historical average remaining gain from Day 14 = 1,255 g.
3. Baseline projection = **380 + 1,255 = 1,635 g**.
4. Revised Day 35 goal = 1,800 g.
5. Gap = **1,635 − 1,800 = −165 g**.

Interpretation: the simple benchmark projects 1,635 g, or 165 g below goal.

During cross-validation, the held-out cycle is excluded before calculating the average remaining gain. This prevents the answer from using its own future result.

Do we still use it?

- **Yes, as a benchmark and possible fallback.**
- **No, not as the live champion while Ridge remains better than the 5% tolerance.**

If future retraining shows Ridge no longer beats it reliably, Canary can safely fall back to this simpler formula.

9. Component 2B — Harvest-recovery model

Business question and target

Question: Given the current daily flock evidence, what final recovery should we expect?

Y: last recorded population divided by beginning population for an eligible historical completed-record cycle.

This is the agreed capstone proxy. It is not a verified harvest-event count.

Evidence and inputs

- **25 independent building-cycle outcomes**
- **5 eligible historical cycles:** 2025-2 through 2026-1
- **122 balanced as-of snapshots**

Selected compact inputs:

- cycle day
- current percentage alive
- latest daily mortality per 1,000 birds
- recent 3-day mortality per 1,000
- mortality trend versus baseline
- latest and cumulative feed per 1,000 birds
- recent temperature
- recent humidity

Raw beginning population and building identity were removed from the selected model. Corrected bodyweight did not materially improve recovery validation and is not in the champion.

Models compared

Candidate	Held-out MAE
Current-survival trend projection	about 3.55 percentage points
Historical mean	about 1.67 points
Ridge with all tested inputs	about 1.50 points
Random Forest	about 1.46 points
Ridge without weight	about 1.34 points
Compact Ridge	**about 1.32 points**

Selection rule and result

Primary metric: cycle-balanced MAE, with a 10% simplicity tolerance for recovery.

Compact Ridge was selected because it had the lowest overall MAE, remained close to the best cycle-balanced candidate, and avoided questionable identity and raw-inventory features.

- Overall MAE: **about 1.32 percentage points**
- RMSE: **about 1.76 points**
- Day 14 MAE: **about 1.43 points**
- Empirical 80% half-width: **about ±2.25 points**

Critical interpretation

Target-side accuracy is about 84%, but that equals the always-below majority baseline. At Day 14, all 25 forecasts were below 95%, including four flocks that later finished at or above 95%.

Therefore:

- useful as a continuous planning estimate
- useful for ranking likely recovery gaps
- **not proven as a classifier of who will hit 95%**

Feature importance

Current survival has the strongest fitted reliance. Recent mortality and feed also contribute. Temperature/humidity and their missing-data indicators appear, but sparse environmental coverage means their importance must not be treated as causal proof.

10. Component 3 — Recommendations

Canary maps the risk pattern, severity, and operational alerts to a deterministic playbook.

Pattern	Plain meaning	Recommended focus
No Material Drift	No major warning in recorded evidence	Continue normal monitoring
Weight Lag Only	Weight is behind the age target	Reweigh; inspect feed, water, access, environment, and uniformity
Survival Concern Only	Survival or mortality is concerning	Reconcile counts; inspect flock, water, feed, litter, and environment
Growth + Survival Drift	Weight and survival are both weak	Conduct a focused flock assessment; escalate if health signs appear
Localized Building Drift	One building is worse than peers	Compare equipment, sensors, feed/water access, litter, and bird distribution
Farm-Wide Drift	Several buildings weaken together	Review shared feed, water, controller, weather, chick-source, or farm-wide changes
Missing or Stale Evidence	Required evidence is absent or old	Collect the missing measurement before a major decision

Urgency follows the risk label:

- Low: routine monitoring
- Medium: within 24 hours
- High: current shift
- Critical: immediate inspection and escalation where appropriate

Recommendations are management and inspection guidance. They do not diagnose disease, prescribe medication, or automatically change house settings.

11. Environment and actionability

Environmental alerts are separate from risk points but appear in the building investigation view.

Current provisional temperature references:

- Days 1–7: 29–33°C
- Days 8–14: 25.5–29.5°C
- Days 15–21: 26–28.5°C

No automatic temperature alert is issued after Day 21 until the farm approves a later-age range. Humidity and feed alerts are also provisional. Water and THI require reliable inputs and approved formulas.

Canary should say "temperature is above the approved range; inspect ventilation/cooling and verify the sensor," not automatically "lower temperature by 3°C." A specific adjustment must depend on the verified sensor, bird behavior, equipment, housing, weather, and farm protocol.

12. Business value estimator

For a recovery shortfall:

Birds at risk = beginning population × recovery gap

Gross revenue at risk = birds at risk × assumed sale weight × PHP per kg

Example:

- beginning population = 10,000 birds
- projected recovery = 93%
- goal = 95%
- gap = 2 percentage points
- birds at risk = **10,000 × 0.02 = 200**
- assumed sale weight = 2.0 kg
- price = PHP 120/kg
- revenue at risk = **200 × 2.0 × 120 = PHP 48,000**

This is a scenario estimate, not guaranteed recoverable profit. It excludes costs and assumes recovery improvement is achievable.

13. Expected panel questions

"Why not train one model per building?"

Each building has too few completed outcomes. Canary pools all eligible buildings and cycles, then holds out whole cycles. A per-building model would be unstable and impossible to validate credibly.

"Why are there 124 weight rows but only 31 outcomes?"

Each building-cycle is recreated at four decision checkpoints. Cross-validation groups the entire cycle so those related rows never split between train and test.

"Why Ridge rather than Random Forest?"

Ridge had the best cycle-balanced error for Day 35 weight, is more stable on a small dataset, and is easier to explain. Random Forest was close but not better. Gradient boosting was worse.

"Why not just use average daily gain?"

Straight-line ADG assumes the recent growth rate continues unchanged. In held-out testing its MAE was about 432 g, far worse than Ridge at about 172 g.

"Why keep historical remaining gain?"

It is transparent, strong, and easy to audit. It is the benchmark the ML model must beat. Ridge beat it beyond the 5% tolerance, so Ridge is champion today.

“Can we trust the recovery model?”

Trust it as a limited-data continuous estimate with about 1.3 points MAE—not as proof that a building will hit 95%. It did not beat the majority baseline for target classification.

“Do temperature and humidity drive the forecast?”

They are recovery-model inputs and operational alerts, but sparse data prevents causal claims. Use them to guide inspection, not to claim a guaranteed intervention effect.

“Is the app production ready?”

No. It is capstone-prototype ready after final farm validation. Production use still needs stronger data governance, confirmed harvest events, threshold approval, secure deployment, monitoring, and more cycles.

14. What we can defend strongly

- one-row-per-building-day data preparation
- correct handling of Zone A/B records without double-counting production
- corrected checkpoint-weight coverage
- rules-based score traceability
- cycle-held-out model validation
- three ML candidates plus transparent baselines for Day 35 weight
- compact, leakage-safe recovery model
- visible uncertainty and limitations
- deterministic recommendation mapping

15. What we must not overclaim

- risk score is not a probability
- feature importance is not causality
- recovery target classification is not proven
- five Day 35 target hits are a small group
- unweighted zone averaging assumes equal relevance of Zones A and B
- the recovery target is a last-recorded proxy, not a verified harvest-event result
- recommended actions support inspection; they do not guarantee recovered revenue or target attainment

16. Final one-sentence defense

Project Canary converts daily building data into transparent operational priorities, leakage-safe outcome forecasts, and practical inspection guidance, while clearly separating what the evidence supports from what still requires farm validation.